WORKED SOLUTION

STEP II 1990, Question 8

Pure • Integration by Parts

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q8

The key idea

Every term of the differential equation is, or nearly is, an exact derivative. Since y and y' take the same values at $t = 0$ and $t = 1$, those exact derivatives integrate to zero. Each relation comes from multiplying the equation by a suitable function and then integrating by parts until every boundary term vanishes. Keep a list of the facts that kill boundary terms: $x(0) = 0$, $y(0) = y(1)$, $y'(0) = y'(1)$, and (once we have proved it) $x(1) = 0$.

Solution

The first identity

Since $x' = y$, the integrand $xy = xx'$ is the derivative of $\frac{1}{2}x^2$. So

$$\int_0^1 x(t)\, y(t)\, \mathrm{d}t = \Big[\tfrac{1}{2}x(t)^2\Big]_0^1 = \tfrac{1}{2}[x(1)]^2,$$

because $x(0) = \int_0^0 y(u)\, \mathrm{d}u = 0$. This is the required result.

Showing $x(1) = 0$

Write the middle term of $(*)$ as an exact derivative:

$$(y^2 - 1)\, y' = \frac{\mathrm{d}}{\mathrm{d}t}\Big(\tfrac{1}{3}y^3 - y\Big).$$

Now integrate $(*)$ from 0 to 1 term by term:

$$\Big[y'\Big]_0^1 + \Big[\tfrac{1}{3}y^3 - y\Big]_0^1 + \int_0^1 y\, \mathrm{d}t = 0.$$

The first bracket is $y'(1) - y'(0) = 0$, and the second is zero because $y(1) = y(0)$. The integral left over is $x(1)$, by the definition of x. Hence $x(1) = 0$.

The relation $\int y^2 = \frac{1}{3}\int y^4$

Multiply $(*)$ by x and integrate over $[0,1]$. There are three integrals, and we handle them one at a time.

The last term. By the first identity together with $x(1) = 0$,

$$\int_0^1 x\,y\,\mathrm{d}t = \tfrac{1}{2}[x(1)]^2 = 0.$$

The y'' term. We integrate by parts, differentiating x (so that $x' = y$):

$$\begin{aligned}\int_0^1 x\,y''\,\mathrm{d}t &= \Big[x\,y'\Big]_0^1 - \int_0^1 y\,y'\,\mathrm{d}t \\ &= x(1)y'(1) - x(0)y'(0) - \Big[\tfrac{1}{2}y^2\Big]_0^1 = 0,\end{aligned}$$

since $x(1) = x(0) = 0$ and $y(1) = y(0)$.

> **Ask yourself** How should we integrate $x\,(y^2-1)\,y'$ by parts? Which factor do we integrate?

The middle term. Integrate the factor $(y^2-1)y'$, which we already know is the derivative of $\frac{1}{3}y^3 - y$, and differentiate x:

$$\begin{aligned}\int_0^1 x\,(y^2-1)\,y'\,\mathrm{d}t &= \Big[x(\tfrac{1}{3}y^3 - y)\Big]_0^1 - \int_0^1 y(\tfrac{1}{3}y^3 - y)\,\mathrm{d}t \\ &= 0 - \tfrac{1}{3}\int_0^1 y^4\,\mathrm{d}t + \int_0^1 y^2\,\mathrm{d}t.\end{aligned}$$

The boundary term vanishes because x is zero at both ends.

Adding the three pieces gives zero:

$$\int_0^1 y^2\,\mathrm{d}t - \tfrac{1}{3}\int_0^1 y^4\,\mathrm{d}t = 0, \qquad \text{that is,} \qquad \int_0^1 [y(t)]^2\,\mathrm{d}t = \tfrac{1}{3}\int_0^1 [y(t)]^4\,\mathrm{d}t,$$

as required.

The relation $\int y'^2 = \int y^2$

> **Ask yourself** We want an integral of y'^2. Which multiplier will produce one?

Integrating $y\,y''$ by parts produces $-\int y'^2$, so this time we multiply $(*)$ by y and integrate. Taking the terms in turn:

$$\int_0^1 y\,y''\,\mathrm{d}t = \Big[y\,y'\Big]_0^1 - \int_0^1 y'^2\,\mathrm{d}t = -\int_0^1 y'^2\,\mathrm{d}t,$$

$$\int_0^1 (y^3 - y)\,y'\,\mathrm{d}t = \Big[\tfrac{1}{4}y^4 - \tfrac{1}{2}y^2\Big]_0^1 = 0.$$

The boundary term $y(1)y'(1) - y(0)y'(0)$ is zero because both y and y' repeat. The second line vanishes because $y(1) = y(0)$. What remains is

$$-\int_0^1 y'^2\,\mathrm{d}t + \int_0^1 y^2\,\mathrm{d}t = 0, \qquad \text{so} \qquad \int_0^1 [y'(t)]^2\,\mathrm{d}t = \int_0^1 [y(t)]^2\,\mathrm{d}t.$$

Checks

The first identity holds for any y. With $y = \cos 3t + t^2 - 0.7e^t$, quadrature gives $\int_0^1 xy\,\mathrm{d}t \approx 0.338191$, which equals $\frac{1}{2}x(1)^2$.

A non-trivial solution of $(*)$ that repeats after time 1 doesn't exist (see below). However, the same arguments work word for word on $[0, T]$ for any solution that repeats after time T. We integrated $(*)$ numerically onto its periodic orbit, which has $T \approx 6.6633$. Over one period we found $\int y \approx 10^{-9}$, $\int y^2 \approx 13.72222$, $\frac{1}{3}\int y^4 \approx 13.72222$ and $\int y'^2 \approx 13.72228$, which agree to within the step-size error.

Common slips

- Forgetting that $x(1) = 0$ is needed in the third part. Without it, neither $\int xy$ nor the boundary terms vanish.
- In the middle term, integrating x instead of $(y^2 - 1)y'$. Using $x' = y$ is what makes the integral collapse to powers of y.
- Saying a boundary term is zero "because $y(0) = 0$". We aren't told that. The boundary terms vanish because the values at 0 and 1 are equal.

Beyond STEP

Equation $(*)$ is the van der Pol oscillator, a classic model of a self-sustaining electrical circuit. It has a single periodic orbit, and the relations you proved are energy balances around one cycle. They also reveal a surprise. Since $\int_0^1 y = 0$, the Wirtinger (or Poincaré) inequality from Fourier series gives $\int_0^1 y'^2 \geq 4\pi^2 \int_0^1 y^2$. Together with the last relation, this forces $y \equiv 0$. So the only solution that repeats after time 1 is the trivial one, because the true period is about 6.66. For more on periodic orbits and limit cycles, see Boyce and DiPrima, *Elementary Differential Equations*.

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